Taxi-System Simulation and Statistical Output Analysis

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***Abstract*—This report presents two connected simulation laboratories. The first develops a discrete-time taxi model with spatial demand, app dispatching, taxi stands and fleet relocation, and studies daily demand and supply shocks. The second introduces constant-input experiments, an online warm-up detector and independent-replication confidence intervals. Rain-induced slowdowns approximately double app waiting relative to a demand-only rain control. Combined stress leaves a substantial backlog. In the stationary-input baseline, mean app waiting is 6.524 minutes with a nominal 95% interval of [6.300, 6.747]. Longer measurements improve precision, whereas greater demand increases both waiting and uncertainty. Assumptions, incomplete observations and the heuristic nature of transient detection are explicitly distinguished.**

I. Problem Overview

The first laboratory asks for a tractable model of passenger generation, closest-idle-taxi dispatch, trip execution, return to stands and optional fleet rebalancing. Its purpose is to identify relevant parameters and performance indicators and explain their effects through scenario experiments. The second laboratory adds online detection of the end of the initial transient, reliable output inference and a study of factors affecting accuracy.

These objectives require two different experimental settings. Daily rush hours, scheduled arrivals and events produce *non-stationary inputs*; their 24-hour outcomes are terminating-simulation results. Constant-input runs are used separately to investigate steady-state performance. Constant inputs are necessary for this analysis but do not guarantee stable queues.

II. Model Design and Assumptions

A. Spatial Mobility and Service Rules

The following assumptions define the model and delimit its interpretation.

1. The city is a fixed 10×10 km square divided into four equal zones. Urban origins are uniform within each zone; destination zones follow a fixed origin–destination probability matrix, followed by uniform sampling within the destination zone. This captures spatial heterogeneity, but not directional changes in commuting patterns.
2. Each passenger requests one vehicle, using app/phone with probability $p_{\text{app}} = 0.70$ or walking to the nearest stand. Modes and destinations are exogenous. Walking speed is constant; passengers do not abandon, switch mode or change stand after joining. Passenger queues have no capacity limit.
3. Taxis are homogeneous, carry one passenger and serve one trip at a time. There are no shifts, breaks, fares, fuel constraints or ride sharing, and no separate boarding or alighting dwell time. Only taxis parked in state `idle` are dispatchable; returning or rebalancing taxis cannot accept app requests. App requests are processed FIFO and assigned to the nearest idle taxi. Stand queues are also FIFO.
4. Eight ordinary stands have ten parking spaces each. The station and airport add stands with 14 and 18 spaces. A taxi returns to the nearest stand with currently free space; places are not reserved en route, so a taxi may be redirected on arrival. Waiting stand passengers are served before parking.
5. Every run starts with no passengers and taxis distributed round-robin over stands, respecting capacity. Speeds depend on prescribed conditions, not an endogenous traffic network. All numerical values are illustrative, without calibration to a real city.

For positions x, y, the road-distance approximation and walking time are

$$d(x,y) = \kappa \|x-y\|_2, \qquad t_{\text{walk}}(x,s) = \frac{60\|x-s\|_2}{v_w}, \quad (1)$$

where $\kappa = 1.20$, $v_w = 4.8$ km/h and times are in minutes. Taxis move along straight segments, with progress scaled to the road distance. In an interval of h minutes, the road distance travelled is

$$\Delta d = \min\left\{ d_{\text{remaining}}, \frac{v(t,x)h}{60} \right\}. \quad (2)$$

The effective speed is the time-of-day baseline multiplied by weather, transit, event and hub-congestion factors, with a floor of 6 km/h. The baseline is 20 km/h in rush hours, 32 at night and 28 otherwise. These factors are exogenous supply reductions rather than a traffic-flow model. In the rain experiments, the weather factor is $\max(1-0.30r, 0.40)$; the demand-only control sets the coefficient 0.30 to zero. The transit speed factor is $\max(1-0.18(1-a_z), 0.50)$. A strike sets availability to 0.25 between 06:00 and 22:00.

B. Demand and Fleet Management

Ordinary demand in zone z is approximated on the simulation grid by independent Poisson counts:

$$N_z(t,t+h) \sim \text{Poisson}(\lambda_z(t)h/60), \tag{3}$$

$$\lambda_z(t) = \lambda_z^0 g_h(t) g_d(t) m(1+0.35r) \times [1+(1-a_z(t))]\, e_z(t). \tag{4}$$

The four base rates λ_z^0 are 5, 6, 4 and 5 passengers/hour. g_h and g_d encode hour and weekday, m scales urban demand, $r \in [0,1]$ is rain intensity and $a_z \in [0,1]$ is transit availability. The factor e_z equals 0.70 in the event zone during the event and one otherwise.

At each scheduled hub arrival, the model samples $L \sim \text{Poisson}(\ell)$ travellers and $N_h \mid L \sim \text{Binomial}(L,q)$ taxi users. The station uses $(\ell,q) = (70, 0.08)$ and 13 daily arrivals; the airport uses $(55, 0.16)$ and nine arrivals. Their app probabilities are 0.45 and 0.60. Origins have small Gaussian perturbations around the terminal, clipped to the city. Four percent of ordinary urban trips target a hub when hubs are enabled.

Event demand is explicitly directional: passengers travel to the venue before the event and leave it afterwards. For inbound volume V_{in}, outbound volume V_{out}, pre-window length L_{in} and post-window length L_{out}, the per-minute rates are

$$\eta_{\text{in}}(s) = \frac{2V_{\text{in}}s}{L_{\text{in}}^2}, \qquad 0 \le s < L_{\text{in}}, \tag{5}$$

$$\eta_{\text{out}}(u) = \frac{V_{\text{out}} e^{-u/\tau}}{\tau(1-e^{-L_{\text{out}}/\tau})}, \qquad 0 \le u < L_{\text{out}}. \tag{6}$$

Counts are Poisson with mean equal to rate times step length. The tested event lasts 20:00–22:00, with attendance 1,100, taxi shares 0.07 inbound and 0.11 outbound, $L_{\text{in}} = 120$, $L_{\text{out}} = 90$ and $\tau = 28$ minutes. Local origins or destinations are perturbed around the venue and clipped to city bounds.

Every 30 minutes, rebalancing computes desired idle counts proportional to forecast zone demand, including upcoming hub and outbound-event requests. At most $\lfloor 0.25 I(t) \rfloor$ idle taxis are moved from surplus to deficit zones, where $I(t)$ is the number idle. This is a simple zone-level rule, not an optimizer of the current queues at individual stands.

C. Time Convention and Performance Indicators

The nominal resolution is one minute. Generation, app dispatch and rebalancing use the state at the beginning of each interval; taxis and walkers then advance. Positive-duration movement completions are recorded at its end. Zero-distance access is immediate, and the final interval is shortened to the requested horizon. Time discretization remains a source of approximation.

Let r_i, s_i, p_i and d_i denote request, stand-arrival, pickup and dropoff times. For the observed sets of app passengers already boarded ($\mathscr{A}_b$), stand users who arrived ($\mathscr{S}_a$) and stand users boarded ($\mathscr{S}_b$),

$$\overline{W}_{\text{app}} = \frac{\sum_{i\in\mathscr{A}_b}(p_i - r_i)}{|\mathscr{A}_b|}, \tag{7}$$

$$\overline{T}_{\text{walk}} = \frac{\sum_{i\in\mathscr{S}_a}(s_i - r_i)}{|\mathscr{S}_a|}, \qquad \overline{W}_{\text{stand}} = \frac{\sum_{i\in\mathscr{S}_b}(p_i - s_i)}{|\mathscr{S}_b|}. \tag{8}$$

Total access is $p_i - r_i$, including walking for stand users. Completion and fleet-state shares over horizon T, with F taxis, are

$$C = \frac{N_{\text{completed}}}{N_{\text{generated}}}, \qquad U = \frac{\sum_f T_f^{\text{occupied}}}{FT}, \qquad A = \frac{\sum_f T_f^{\text{idle}}}{FT}. \tag{9}$$

Empty-running time and empty-distance share are reported separately. Availability A is a time–fleet fraction, not the probability of immediate assignment. Undefined means for empty populations are recorded as missing, not zero.

Daily waiting means are conditional on observed boarding, including passengers still travelling. The model also reports backlogs and lower-bound means over all arrivals. For the set $\mathscr{A}$ of all app requests observed by T,

$$\overline{L}_{\text{app}}(T) = \frac{1}{|\mathscr{A}|}\left[\sum_{i\in\mathscr{A}_b}(p_i - r_i) + \sum_{i\in\mathscr{A}\setminus\mathscr{A}_b}(T - r_i)\right]. \tag{10}$$

An unserved user's final waiting time is still unknown. Thus unfinished requests are not silently treated as served or as zero-wait observations.

TABLE I
Daily results: means over five replications. Times are minutes.

Scenario	App wait	Walk	Stand wait	C (%)
Baseline	6.08	12.08	22.56	98.8
No hubs	5.11	18.65	2.81	99.0
Rain: demand only	6.52	12.36	32.24	98.1
Rain: full effect	13.35	12.36	37.55	97.7
Transit strike	12.73	13.54	22.22	98.9
Evening event	11.07	13.06	25.70	96.8
Combined stress	151.13	15.23	28.16	71.9

III. Laboratory 1: Daily Scenario Results

Seven 24-hour scenarios were run with $F = 40$ and five replications each, using seeds 1000–1004. The baseline is Wednesday with hubs and rebalancing enabled. The scenarios change hub presence, rain, transit service or the evening event; combined stress also uses Friday demand. All tables average per-replication statistics, rather than pooling passengers across runs.

Table I and Fig. 1 show that demand and effective supply must be considered together. The two rain scenarios generate exactly the same requests for matched seeds: their mean demand is 789 users/day. Adding the weather speed penalty increases app waiting from 6.515 to 13.352 minutes, approximately 105%, and lowers the idle share from 65.1% to 50.3%. This comparison isolates the additional effect of slower circulation; the baseline generates only 619.8 users/day.

The event produces an average 198.4 event requests and reduces completion to 96.8%. Combined stress generates 1,433.4

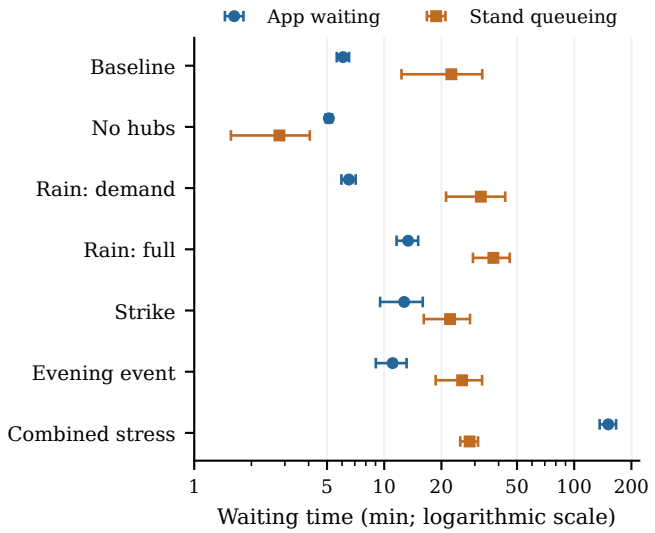


Fig. 1. Daily waiting means with sample standard deviations across five runs. The bars are descriptive variability, not confidence intervals. Waiting is conditional on boarding; the logarithmic axis keeps combined stress visible.

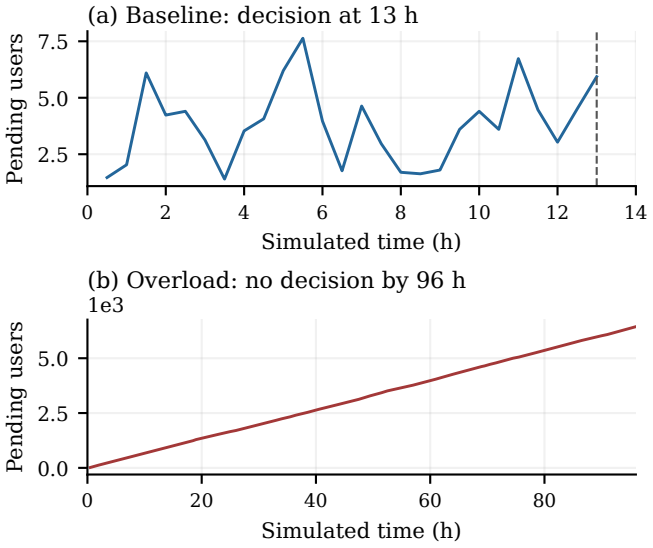


Fig. 2. Pending-user batch averages from pilot 0 (seed 3481032016). The baseline pilot stops at its 13-hour decision. Under overload, pending users continue accumulating throughout the 96-hour screening horizon.

requests, with average counts of 353.0 app users not yet boarded, 28.6 stand users not yet boarded and 21.4 passengers still travelling at midnight. The 71.9% completion rate therefore does not imply that the other requests were abandoned. Its 151.131-minute boarded-app mean is complemented by an all-app-arrival lower-bound mean of 158.315 minutes, exposing the unresolved backlog.

Baseline occupied and idle shares are 16.3% and 71.6%; under combined stress they become 42.1% and 24.5%. Residual idle time can coexist with long queues because of spatial mismatch, varying load and the restriction that returning taxis are unavailable. Stand delays can likewise remain high when other stands contain idle taxis.

The no-hub comparison changes demand, stand locations, total parking capacity (112 versus 80), initial taxi positions, some destinations and local congestion. It compares two configurations, not the isolated causal effect of arrival bursts. Its longer walking mean also reflects the absence of short terminal walks. With only five runs, these daily experiments are descriptive and do not establish precise steady-state effects or real-city predictions.

IV. Laboratory 2: Stationary Output Analysis

A. Constant-Input Experiments

The stationary mode fixes $g_h = g_d = 1$ and the base taxi speed at 28 km/h. Hubs, events, temporary disruptions and periodic rebalancing are disabled. Baseline urban demand is consequently 20 users/hour, with $F = 20$, eight stands and the same spatial rules as before. Rain is zero and transit availability one. This is a different configuration from Laboratory 1, not a warm-up correction applied to its non-stationary daily scenarios.

B. Online Transient Detection

The monitored state is $X(t) = (Q(t), U_{\text{inst}}(t), G(t))$: the number of not-yet-boarded users including walkers, the instantaneous occupied fleet fraction, and the mean age of those pending requests (zero if none). Monitoring current state, rather than cumulative performance averages, helps expose persistent growth or starvation.

Non-overlapping 30-minute, time-weighted batch averages feed a bounded-memory detector. On the latest $2w$ batches ($w = 12$), let $\bar{X}_j^-$ and $\bar{X}_j^+$ be the means of the older and newer windows, s_j the sample standard deviation across both, and $\hat{\beta}_j$ the least-squares slope against batch index. For each component,

$$S_j = a_j + 0.5 s_j, \qquad D_j = \frac{|\bar{X}_j^+ - \bar{X}_j^-|}{S_j}, \tag{11}$$

$$L_j = \frac{|\hat{\beta}_j|(2w-1)}{2S_j}. \tag{12}$$

The absolute tolerances are $a = (1.5, 0.05, 5)$ in users, fleet fraction and minutes. A check passes if $D_j \le 1$ and $L_j \le 1$ for every component. Three consecutive passes trigger a decision; the earliest possible decision is 13 simulated hours. Only already observed data are used. These thresholds are practical tolerances, not significance levels or a formal stationarity test.

Three independent pilot runs determine the common deletion time $B = \max_{k=1,2,3} \hat{B}_k$. If any pilot fails to detect stability within 96 hours, the configuration is marked as undetected and receives no steady-state interval. Fig. 2 contrasts a representative baseline trajectory with persistent growth under overload. Non-detection alone is not a mathematical proof of instability.

C. Independent Replications and Confidence Intervals

Production uses new random streams, independent of the pilots. Each of $R = 20$ replications starts from the prescribed empty state, discards the same B and measures a fixed $H = 24$ hours. Separating pilot selection from production avoids choosing the deletion time from the observations used for inference [1].

For waiting metrics, the cohort contains every arrival in $[B, B+H)$. Normal arrivals continue while this cohort is followed to pickup, for at most another 48 hours. Stopping new

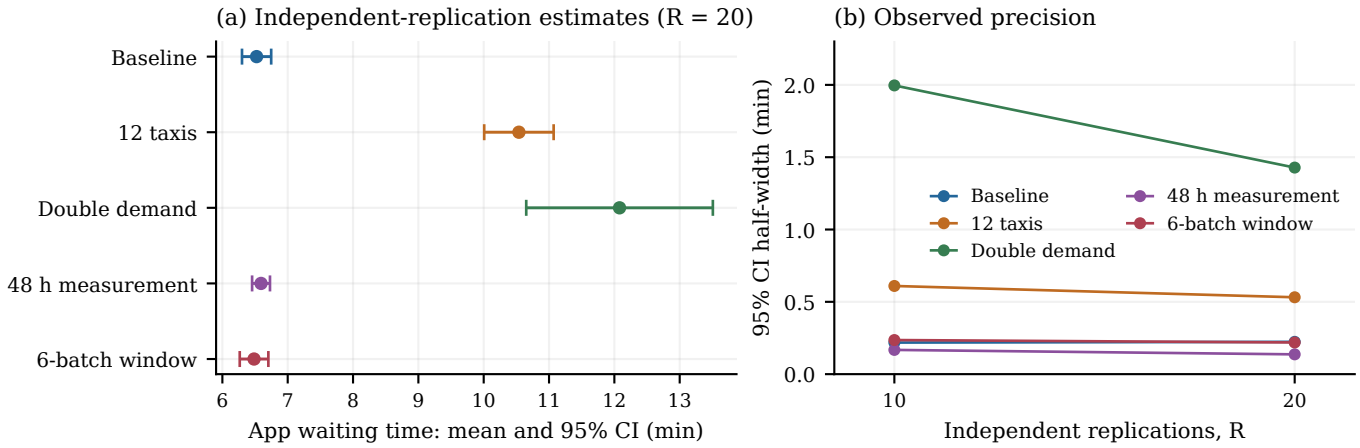


Fig. 3. Stationary-input app waiting. (a) Nominal 95% Student-t intervals across 20 independent production replications. (b) CI half-width for two predeclared sample sizes, using the same replication prefix. Configurations without detected stability are excluded from interval estimation, not silently counted as successful runs.

arrivals would change the service environment. A timeout invalidates the measurement; failed replications are not dropped to manufacture an interval. No such timeout occurred in the reported production runs. Fleet shares use counter differences over $[B, B+H)$ only, excluding follow-up time.

Let Y_r be one replication's mean for a specified KPI. With $\bar{Y}$ the average across replications and s_Y their sample standard deviation,

$$\bar{Y} = \frac{1}{R}\sum_{r=1}^{R} Y_r, \qquad s_Y^2 = \frac{\sum_{r=1}^{R}(Y_r - \bar{Y})^2}{R-1}, \tag{13}$$

$$\mathrm{CI}_{1-\alpha} = [\bar{Y} - h_R, \bar{Y} + h_R], \quad h_R = t_{1-\alpha/2,R-1}\frac{s_Y}{\sqrt{R}}. \tag{14}$$

Student's critical value [2] is used with $\alpha = 0.05$. The sample size and observation duration are fixed before production; the first ten and all twenty replications form two predeclared precision comparisons. Individual minutes or passenger waits are not treated as independent samples, since queueing induces serial dependence. The interval is approximate for replication means and remains sensitive to residual initial bias and non-normality.

The estimand is the expected per-replication window mean, approximating a steady-state KPI when deletion is adequate. It is not a pooled mean across all customers, an individual passenger prediction interval, or a simultaneous 95% statement about every configuration. Seeds are reused across configurations, but no paired-difference intervals are claimed.

V. Accuracy and Parameter Sensitivity

Accuracy involves both variance and initial-condition bias:

$$\mathrm{MSE}(\bar{Y}) = \mathrm{Var}(\bar{Y}) + \mathrm{Bias}(\bar{Y})^2. \tag{15}$$

The exact taxi steady-state mean is unknown, so interval widths measure precision directly, while deletion sensitivity probes potential bias without quantifying it exactly. Reference-model coverage is checked separately.

The five screened configurations yielded 100 complete production replications; all three overload pilots failed screening. The baseline app-wait estimate is 6.524 minutes, with CI [6.300, 6.747]. Mean total access is 14.612 minutes [13.676, 15.548], while occupied share is 0.2257 [0.2215, 0.2299]. Table II and Fig. 3 address accuracy through the following changes.

TABLE II
Stationary study design and observed common deletion time B.

Configuration	F	m	H (h)	w	B (h)
Baseline	20	1	24	12	13.0
Fewer taxis	12	1	24	12	15.0
Higher demand	20	2	24	12	19.0
Longer measurement	20	1	48	12	13.0
Shorter window	20	1	24	6	14.5
Overload	4	4	24	12	—

Fleet and demand. Reducing the fleet from 20 to 12 raises estimated app waiting to 10.539 minutes [10.008, 11.071]. Doubling demand to 40 users/hour with 20 taxis raises it to 12.079 [10.651, 13.508]. The app-wait relative half-width $h_R/|\bar{Y}|$ rises from 3.4% in the baseline to 5.0% and 11.8%, respectively. Thus increased load affects precision as well as service level: the same replication budget gives less precise estimates in the more variable high-demand system. Its deletion time also increases from 13 to 19 hours.

Measurement duration. Extending useful observation from 24 to 48 hours reduces the app-wait half-width from 0.223 to 0.137 minutes, approximately 38.7%. The corresponding estimate, 6.592 [6.455, 6.729], is compatible with the baseline interval. Total-access half-width also falls, from 0.936 to 0.557 minutes. Longer windows improve averaging here but cannot, by themselves, prove the absence of residual initial bias.

Replication count. At fixed variance, Eq. (14) suggests an approximately $R^{-1/2}$ reduction in uncertainty. In the higher-demand case, increasing R from ten to twenty reduces half-width from 1.996 to 1.428 minutes. The baseline instead changes slightly from 0.218 to 0.223: its estimated between-replication standard deviation increases from 0.305 to 0.477. The expected precision trend is not a guarantee of monotonic improvement in one realized sample. There is no adaptive stopping at a desired interval width.

Detector window. With $w = 6$, the first pilot decides at seven hours, but the conservative maximum across pilots is 14.5 hours, versus 13 with $w = 12$. Shorter windows therefore need not shorten the final deletion time. The app estimate becomes 6.485 [6.266, 6.704]. Agreement is a useful sensitivity check, not proof that both deletions remove all initial bias.

A. Coverage Check on a Known Queue

Inference is additionally checked on an M/M/1 queue with $\lambda = 0.5$ and $\mu = 1$. Its theoretical mean queueing wait is

$$\mathbb{E}[W_q] = \frac{\lambda}{\mu(\mu - \lambda)} = 1. \tag{16}$$

Workload is initialized from its exact invariant distribution: zero with probability $1 - \rho$, otherwise exponential with rate $\mu - \lambda$, where $\rho = \lambda/\mu$. Subsequent waits obey Lindley's recursion $W_{n+1} = \max\{0, W_n + S_n - A_n\}$ with independent exponential service and interarrival times. This preserves within-replication dependence while removing initial bias from this particular reference test.

Across 300 experiments, each with 20 independent replications of 3,000 customers, 283 nominal 95% intervals contain the true mean: empirical coverage is 94.3%. Its 95% Wilson binomial interval is [91.1%, 96.4%], consistent with the nominal target. The grand mean is 1.0016. This checks the interval method on a solvable dependent-output model; it does not establish coverage for taxi intervals or validate the warm-up heuristic itself.

VI. Verification, Limitations and Conclusions

All 39 automated tests passed. They cover chronological consistency, known trip distances, parking limits, zero relocation, horizon truncation, transient and drifting test signals, a known Student-t interval, stream separation and cohort follow-up. Numerical summaries were cross-checked against the archived replication CSV files. The implementation uses Python 3.12.14, NumPy 2.3.5, pandas 2.2.3 and SciPy 1.17.0. Daily seeds are 1000–1004; the stationary study uses root seed 20261025 and separate pilot/production seed families. The M/M/1 coverage experiment uses seed 71023.

The main limitations are illustrative spatial and demand assumptions, exogenous congestion, one-minute discretization and a basic zone-level relocation policy. Finite-window stability screening can accept a temporary plateau or miss a slow transient; it cannot prove steady state. Intervals quantify sampling precision under the experimental design, not calibration error, discretization bias or model realism. The daily study likewise retains its explicit finite-horizon interpretation and observed backlogs.

Together, the laboratories demonstrate how demand shocks and slower supply create queueing pressure, and why credible output analysis needs more than a single run. Independent production, explicit cohort observation and reported screening failures make uncertainty visible. Longer measurements improve precision in the tested case, while heavier load demands a larger statistical budget. Sensitivity to warm-up settings must accompany narrow intervals.

References

[1] A. M. Law, *Simulation Modeling and Analysis*, 5th ed. McGraw-Hill Education, 2015.

[2] Student, "The probable error of a mean," *Biometrika*, vol. 6, no. 1, pp. 1–25, 1908.